RETAIL SALES ANALYSIS DASHBOARD USING SQL AND POWER BI

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Retail Sales Analysis Dashboard using SQL & Power BI

Introduction

Retail businesses generate large volumes of sales data every day. Analyzing this data helps organizations understand customer behaviour, product performance, regional trends, and overall business performance.

This project focuses on analyzing retail sales data using SQL and Power BI to identify key revenue drivers, customer purchasing patterns, and business opportunities. The analysis aims to support data-driven decision-making and provide actionable recommendations for improving sales performance.

Problem Statement

The objective of this project is to analyze retail sales data to identify top-performing products, customer segments, regions, and sales channels. The analysis helps uncover business insights that can support revenue growth, improve customer engagement, and optimize operational strategies.

Project Objectives

- Analyze retail sales performance.
- Identify top-performing products and categories.
- Understand customer purchasing behavior.
- Evaluate regional and sales channel performance.
- Build an interactive Power BI dashboard for business decision-making.

Dataset Overview

The dataset contains 120,000 retail transactions and includes customer information, product details, sales amounts, transaction dates, regions, sales channels, and payment methods.

Key attributes analyzed include:

• Transaction Date
• Customer ID
• Customer Segment
• Customer Age Group
• Customer Gender
• Product Category
• Product Name

• Region
• Sales Channel
• Quantity Sold
• Sales Amount

Tools Used

• Oracle SQL Developer
• Power BI Desktop
• Microsoft Excel
• Data Cleaning and Transformation
• Data Visualization
• Business Analysis

Data Preparation

- Verified data types.
- Checked for missing values.
- Checked for duplicate records.
- Validated transaction dates.
- Imported cleaned data into Oracle SQL Developer and Power BI.

SQL Queries Performed

Query 1: Total Business Revenue

Purpose: Calculate the total revenue generated by the business.

```
SELECT ROUND(SUM(sales_amount),2) AS
total_revenue
FROM retail_sales_dataset;
```

Script Output x

Task completed in 0.988 seconds

```
TOTAL_REVENUE
-------------
   45357054.5
```

Query 2: Revenue by Product Category

Purpose: Identify the highest and lowest revenue-generating product categories.

```
SELECT category,
SUM(quantity) units_sold
FROM retail_sales_dataset
GROUP BY category
ORDER BY units_sold DESC;
```

Query Result

SQL | All Rows Fetched: 8 in 0.079 seconds

	CATEGORY	UNITS_SOLD
1	Clothing	25223
2	Sports	25183
3	Beauty	25171
4	Groceries	25109
5	Toys	25039
6	Home	24916
7	Electronics	24866
8	Books	24042

Query 3: Top 10 Products by Revenue

Purpose: Identify products contributing the most to overall sales.

```
SELECT *
FROM (
    SELECT product_name,
     ROUND(SUM(sales_amount),2)
revenue
     FROM retail_sales_dataset
     GROUP BY product_name
     ORDER BY revenue DESC
     )
     WHERE ROWNUM <= 10;
```

Query Result

SQL | All Rows Fetched: 10 in 0.028 seconds

	PRODUCT_NAME	REVENUE
1	Bread	2073770.79
2	Lipstick	2063017.35
3	Textbook	1737817.74
4	Smartphone	1631875.41
5	Water Bottle	1605400.95
6	Storage Box	1509331.33
7	Notebook	1478248.61
8	Toy Car	1423744.56
9	Building Blocks	1394417.17
10	Moisturizer	1348009.02

Query 4: Revenue by Customer Segment

Purpose: Analyze revenue contribution from different customer groups.

```
SELECT customer_segment,
ROUND(SUM(sales_amount),2) revenue
FROM retail_sales_dataset
GROUP BY customer_segment
ORDER BY revenue DESC;
```

Query Result × | Script Output × | Task completed in 0.618 seconds

CUSTOMER_SEGMENT	REVENUE
Loyal	11393721.2
New	11331906.6
VIP	11320171.3
Returning	11311255.3

Query 5: Revenue by Region

Purpose: Evaluate regional business performance.

```
SELECT region,
ROUND(SUM(sales_amount),2) revenue
FROM retail_sales_dataset
GROUP BY region
ORDER BY revenue DESC;
```

Script Output × | Query Result × | SQL | All Rows Fetched: 5 in 0.019 seconds

	REGION	REVENUE
1	Central	9237924.11
2	West	9106911.7
3	North	9049438.11
4	South	8991262.89
5	East	8971517.64

Query 6: Monthly Revenue Trend

Purpose: Analyze revenue trends over time and identify seasonal patterns.

```
SELECT EXTRACT(YEAR FROM transaction_date) AS year,
       EXTRACT(MONTH FROM transaction_date) AS month,
       ROUND(SUM(sales_amount),2) AS revenue
FROM retail_sales_dataset
GROUP BY EXTRACT(YEAR FROM transaction_date),
         EXTRACT(MONTH FROM transaction_date)
ORDER BY year, month;
```

Script Output

Task completed in 0.645 seconds

YEAR	MONTH	REVENUE
2024	1	1980147.17
2024	2	1797169.8
2024	3	1885630
2024	4	1862123.73
2024	5	1946801.1
2024	6	1892102.05
2024	7	1907030.93
2024	8	1947600
2024	9	1858013.87
2024	10	1833992.24
2024	11	1955443.63

Script Output

Task completed in 0.645 seconds

YEAR	MONTH	REVENUE
2024	12	1920349.49
2025	1	1925284.04
2025	2	1793455.91
2025	3	1813051.32
2025	4	1862967.35
2025	5	1921171.94
2025	6	1838634.83
2025	7	2007036.02
2025	8	1967685.22
2025	9	1836361.3
2025	10	1949976.97

YEAR	MONTH	REVENUE
2025	11	1783231.12
2025	12	1871794.42

SQL Analysis Summary

The retail sales dataset was analyzed using SQL to gain meaningful insights into business performance. The analysis focused on key areas such as revenue generation, customer behaviour, product performance, regional sales trends, and sales channel effectiveness.

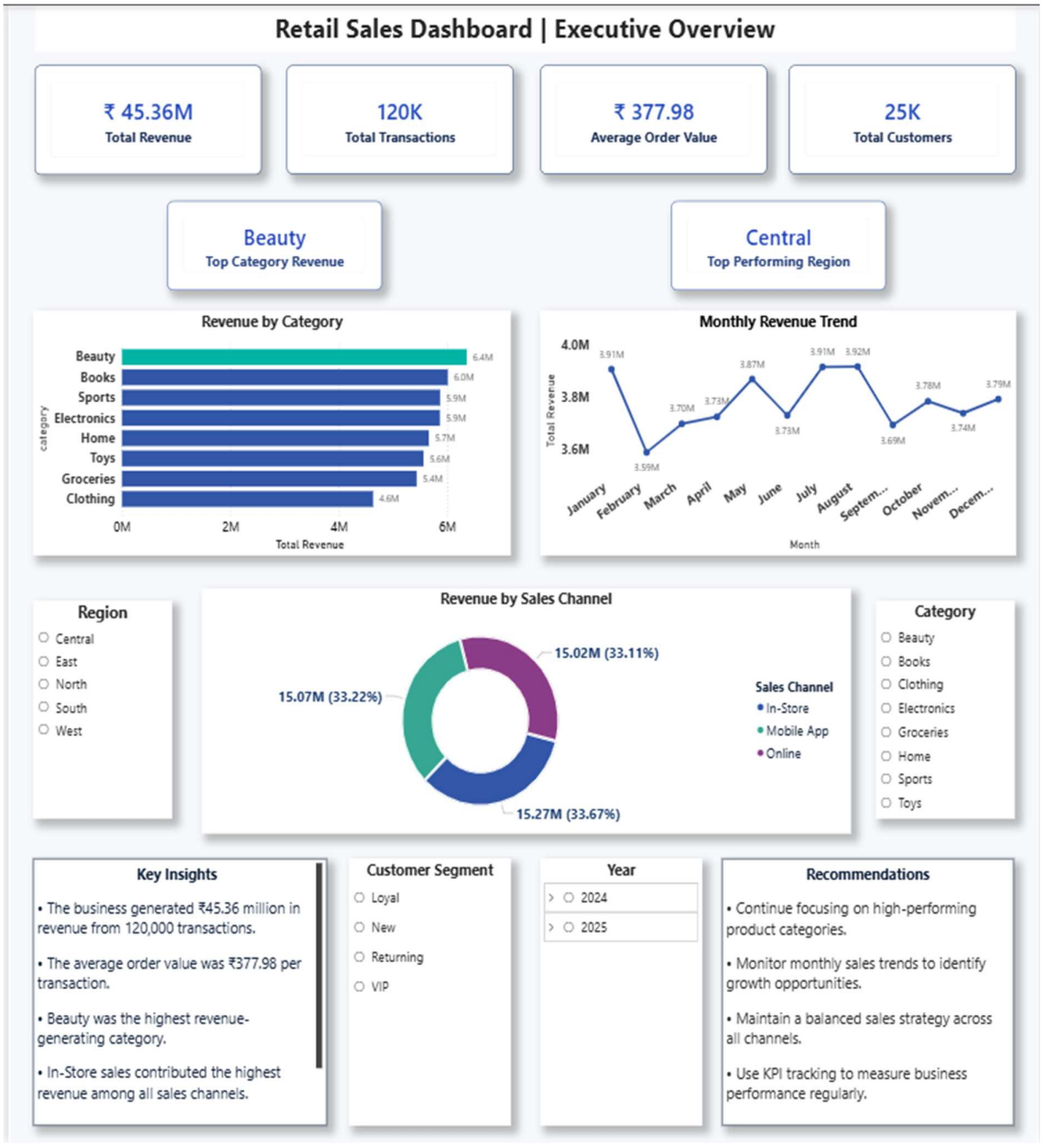
Various SQL techniques, including filtering, aggregation, grouping, sorting, ranking, and trend analysis, were applied to convert raw transactional data into valuable business insights.

The analysis identified Beauty as the highest revenue-generating category, Bread as the top-performing product, the Central region as the best-performing market, and customers aged 35–44 as the highest revenue-contributing age group. These insights formed the basis for the interactive Power BI dashboards and the business recommendations presented in this report.

Dashboard Analysis

Executive Overview

Executive Overview Dashboard showing KPIs, monthly revenue trends, category performance, and sales channel analysis.



Key Insights

- The business generated total revenue of ₹45.36 million from 120,000 transactions.
- The average order value was ₹377.98 per transaction.
- Beauty emerged as the highest revenue-generating category.
- The Central region recorded the strongest sales performance.
- In-Store sales generated the highest revenue among all sales channels.

Recommendations

- Continue investing in high-performing product categories.
- Monitor monthly revenue trends to identify growth opportunities.
- Maintain a balanced sales strategy across all channels.
- Regularly track KPIs to evaluate business performance.

Product Performance

Product Performance Dashboard showing category revenue ranking, top-performing products, product category contribution, units sold by category, and key product performance insights.



Key Insights

• Beauty was the highest revenue-generating product category.

• Bread generated the highest product revenue at approximately ₹2.07 million.

• Several products contributed significantly to overall revenue, indicating a diversified product portfolio.

• Clothing generated the lowest category revenue compared to other categories.

Recommendations

• Increase inventory availability for top-performing products.

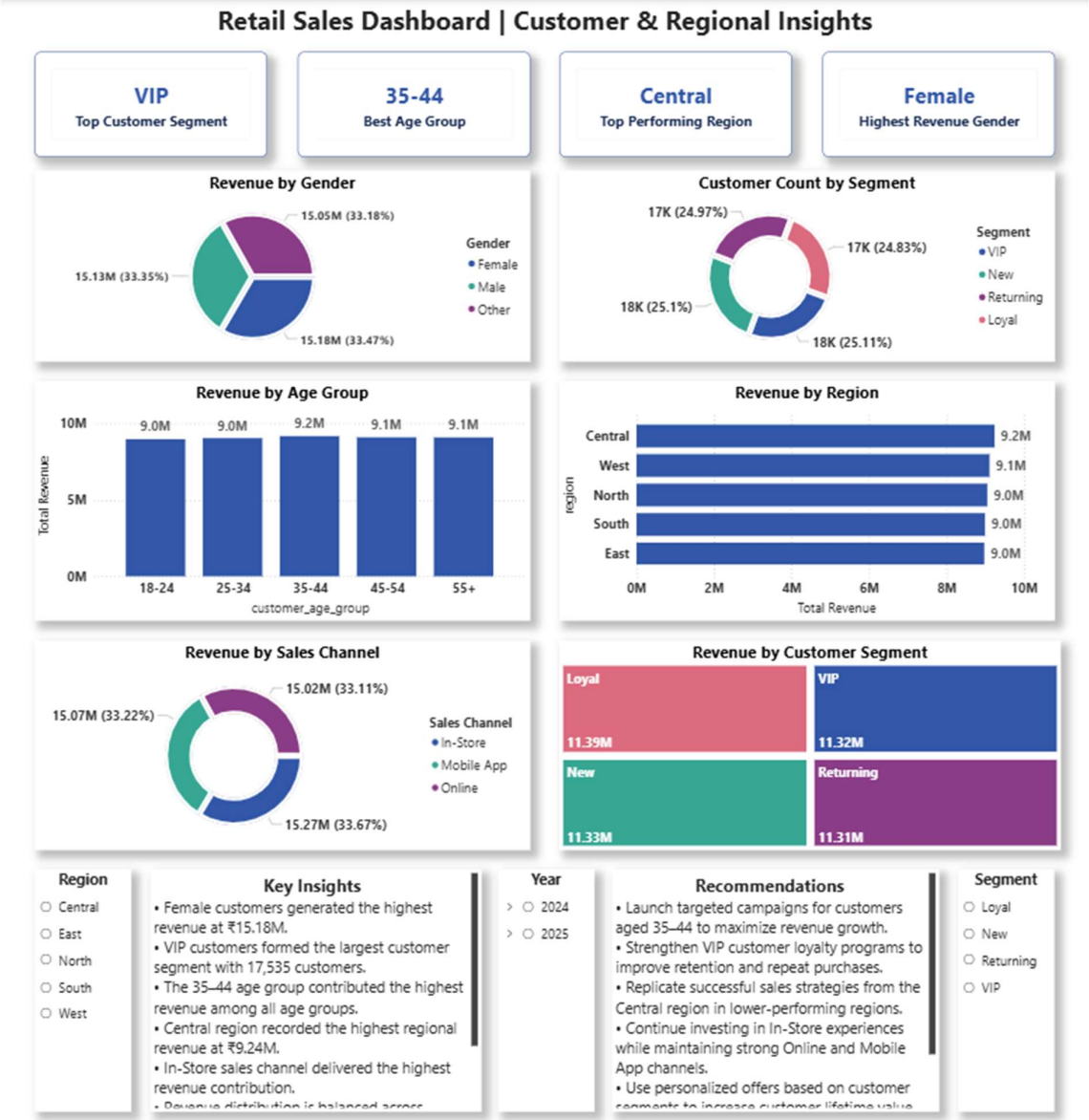
• Promote best-selling products through targeted marketing campaigns.

• Investigate factors affecting lower-performing categories.

• Use product performance insights to improve inventory planning and demand forecasting.

Customer & Regional Insights

Customer & Regional Insights Dashboard showing customer segment analysis, age group and gender-based revenue, regional performance, sales channel analysis, and actionable business insights.



Key Insights

• Female customers generated the highest revenue contribution of approximately ₹15.18 million.

• Customers aged 35–44 contributed the highest revenue among all age groups.

• The Central region generated the highest regional revenue.

• In-Store sales channel delivered the strongest revenue performance.

• Revenue distribution across customer segments remained balanced, reducing dependency on a single customer group.

Recommendations

• Strengthen customer retention programs for valuable customer segments.

• Develop targeted marketing campaigns for customers aged 35–44.

• Replicate successful sales strategies from the Central region in other regions.

• Continue investing in both physical and digital sales channels to maintain balanced growth.

Skills Demonstrated

- SQL (Aggregation, GROUP BY, ORDER BY, Joins, Ranking)
- Data Cleaning
- Data Analysis
- Data Visualization
- Dashboard Design
- KPI Reporting
- Business Intelligence
- Power BI
- Microsoft Excel
- Analytical Thinking

Conclusion

This project demonstrates the complete data analysis workflow, including data exploration, SQL analysis, business insight generation, dashboard development, and recommendation creation. The findings can help stakeholders understand business performance and support strategic decision-making.